

Cancer in the Minnesota Hmong Population

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BACKGROUND. The Hmong are an isolated, agrarian people who settled in the mountainous regions of what today are Vietnam, Cambodia, and Laos. After the Vietnam War, many Hmong were relocated to the U.S. Minnesota has the second largest population (after California) of Hmong individuals. The objective of this study was to examine cancer incidence in this population, because it may indicate areas for targeted surveillance and intervention.

METHODS. The Minnesota Cancer Surveillance System database was screened for Hmong surnames, and proportional incidence ratios (PIRs) were calculated for the period 1988–1999.

RESULTS. Compared with all Minnesotans, the Hmong population had increased PIRs for nasopharyngeal cancer (PIR, 39.39; 95% confidence interval [95% CI], 21.01–66.86), gastric cancer (PIR, 8.70; 95% CI, 5.39–13.25), hepatic cancer (PIR, 8.08; 95% CI, 3.88–14.71), and cervical cancer (PIR, 3.72; 95% CI, 2.04–6.20) and had decreased PIRs for prostate cancer, breast cancer, Hodgkin disease, and melanoma.

CONCLUSIONS. The current observations have implications for cancer control interventions. In particular, an increased incidence of cervical cancer might be addressed in part by targeting culturally sensitive screening programs in the Hmong population. *Cancer* 2003;97:3076–9. © 2003 American Cancer Society.

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The Hmong, an isolated agrarian people from southern China, migrated to the mountainous regions of what today are Vietnam, Cambodia, and Laos.¹ At the conclusion of the Vietnam War, many Hmong were relocated from refugee camps in Thailand to the U.S. and settled in New Jersey, Texas, Wisconsin, California, and Minnesota. Since the 1980s, the Hmong population in Minnesota has increased significantly. Minnesota has the second highest population (after California) of Hmong individuals in the U.S. Although there is some debate regarding the actual number of Hmong individuals who reside in Minnesota (estimates range from 40,000 to 80,000), recent data from the 2000 Census indicate that there are approximately 42,000, more than half of whom live in Saint Paul.²

Because of cultural barriers, high rates of poverty, and language isolation, health status and practices are of interest in this unique migrant population.³ Reports concerning blood pressure, appendicitis, reproductive outcomes, and breast cancer screening practices in the Hmong have been published.^{4–8}

A single study from California examined cancer incidence in the Hmong population in California from 1987 to 1994.⁹ In that report, compared with all races combined, the Hmong had statistically significant higher proportional incidence ratios (PIRs) of gastric, hepatic,

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cervical, nasopharyngeal, and pancreatic malignancies as well as non-Hodgkin lymphoma (NHL). Non-statistically significant elevated PIRs also were observed for leukemia, multiple myeloma, and laryngeal cancer. In contrast, significantly decreased ratios were observed for prostate, breast, kidney, and brain cancers and for melanoma.

MATERIALS AND METHODS

Using the Minnesota Cancer Surveillance System (MCSS), we evaluated cancer incidence in the Hmong population in Minnesota during the 12-year period from 1988 to 1999. The MCSS has been in operation since 1988 and serves as the population-based cancer registry for the state. U.S. 2000 Census data show that approximately 5 million individuals reside in Minnesota, including 88% non-Hispanic whites, 4% African Americans, 3% Asians, 3% Hispanics, and 2% Native Americans/other. An external audit of the MCSS performed in June, 1996 by the North American Association of Central Cancer Registries estimated that the MCSS hospital-based case finding was 99.6% complete for microscopically confirmed cancers and 99.1% complete for all cancers. Screening of the MCSS database resulted in a list of cancers diagnosed between 1988 and 1999 among Minnesota residents, which contained the variables of cancer type (invasive cancers only), gender, age at diagnosis, and an indicator for Hmong surname (Hmong surnames were kindly provided by Dr. Paul Mills, who conducted a similar analysis in California⁹). The final Hmong surname list was more specific than sensitive; surnames were excluded that may have included individuals from other Asian populations.

Because census data on the Hmong population in Minnesota are considered unreliable, it was impractical to calculate cancer incidence rates. Therefore, age-adjusted (20-year groupings: age younger than 20 years, ages 20–40 years, ages 40–60 years, ages 60–80 years, and 80 + years) and gender-adjusted PIRs along with 95% confidence intervals (95% CI) were calculated.^{9–12} Briefly, an observed-to-expected ratio was calculated. The observed number was equal the actual number of Hmong individuals for a particular disease site. The expected number was calculated by multiplying the age specific (20-year groupings) and gender specific numbers of cancers for all sites in the Hmong population by a quotient that was derived as follows: the (age/gender specific) number of the site specific cancer in the entire Minnesota population divided by the corresponding (age/gender) specific number of all

cancers in the Minnesota population. Confidence intervals were calculated using the methods described by Ury and Wiggins.¹²

RESULTS

During 1988–1999, 222,406 incident malignancies were ascertained by the MCSS. A total of 186 malignancies (< 0.1%) occurred in the Hmong population, as determined through surnames used previously.⁹ Eighty-four cancers occurred in males, and 102 cancers occurred in females. The age distribution was as follows: age younger than 20 years, 6%; ages 20–40 years, 23%; ages 40–60 years, 30%; ages 60–80 years, 33%; and age 80 + years (8%).

Age-adjusted and gender-adjusted PIRs for specific malignancies during the period from 1988 to 1999 are shown in Table 1. Other unspecified sites ($n = 13$ malignancies) were not evaluated. Compared with the overall population in Minnesota (who are mostly white), there was a statistically significantly increased PIR in the Hmong population for nasopharyngeal, gastric, hepatic, and cervical malignancies. An elevated PIR also was observed for pancreatic cancer (PIR, 2.45; 95% CI, 0.98–4.96). Statistically significantly decreased PIRs were observed for prostate cancer, and breast cancer, Hodgkin disease, and melanoma.

DISCUSSION

Although most of the data are similar to those reported in the Hmong population in California,⁹ we did not see an increased proportion of NHL. We also observed a statistically significant decreased proportion of Hodgkin disease: The study in California did not report on this malignancy.

The proportional occurrence of several of these malignancies in the Hmong, including nasopharyngeal cancer,¹³ gastric cancer,¹⁴ liver cancer,¹⁵ breast cancer,^{16–18} and prostate carcinoma,^{16–18} is in agreement with findings in other Asian populations. It is noteworthy that gastric, cervical, nasopharyngeal, and hepatic malignancies all are associated with infectious exposures (*Helicobacter pylori*, human papilloma virus, Epstein–Barr virus, and hepatitis B virus, respectively) that are endemic in certain parts of Asia. Thus, it is not surprising that the proportion of malignancies would be higher in Hmong Minnesotans compared with all Minnesotans.

There are some limitations to the approach used. PIRs were calculated rather than direct incidence rates. A limitation of using PIRs is that proportionate excesses for some sites will be accompanied by proportionate deficits in other sites. It was not possible to

TABLE 1
Age-Adjusted and Gender-Adjusted Proportional Incidence Ratios of
Malignancies in the Hmong Population in Minnesota (1988–1999)

Site	Observed	Expected	PIR	95% CI
Lip	1	0.97	1.03	0.04–5.09
Salivary gland	1	0.73	1.38	0.06–6.83
Gum	1	0.88	1.13	0.05–5.61
Nasopharynx	13	0.33	39.39 ^a	21.01–66.86
Tonsil	1	0.47	2.14	0.09–10.59
Nasal	2	0.42	4.75	0.54–16.10
Stomach	21	2.41	8.70 ^a	5.39–13.25
Intestine	1	0.8	1.24	0.05–6.17
Colon	11	13.9	0.79	0.40–1.40
Rectum	7	5.43	1.29	0.52–2.61
Liver	10	1.24	8.08 ^a	3.88–14.71
Gallbladder	2	0.42	4.78	0.55–16.19
Pancreas	7	2.86	2.45	0.98–4.96
Lung	18	18.59	0.97	0.57–1.52
Soft tissue	1	2.86	0.35	0.01–1.73
Prostate	3	19.26	0.16 ^a	0.03–0.44
Penis	1	0.16	6.07	0.24–30.10
Urinary	4	6.84	0.58	0.16–1.45
Kidney	5	5.16	0.97	0.31–2.21
Eye	1	0.65	1.54	0.06–7.64
Brain	3	6.83	0.44	0.09–1.23
Thyroid	7	6.1	1.15	0.46–2.32
Melanomas	2	10.69	0.19 ^a	0.02–0.63
Other skin	1	2.36	0.42	0.02–2.11
Breast	11	32.98	0.33 ^a	0.17–0.59
Uterine cervix	14	3.76	3.72 ^a	2.04–6.20
Uterine corpus	4	5.9	0.68	0.18–1.68
Ovary	4	5.05	0.79	0.21–1.96
Hodgkin	1	5.16	0.19 ^a	0.01–0.96
Non-Hodgkin	6	10.53	0.57	0.21–1.22
Myeloma	1	1.73	0.58	0.02–2.86
ALL	2	2.95	0.68	0.08–2.30
AML	4	1.89	2.11	0.57–5.24
CML	2	1.14	1.75	0.20–5.94

PIR: proportional incidence ratio; 95% CI: 95% confidence interval; ALL: acute lymphocytic leukemia; AML: acute myeloid leukemia; CML: chronic myeloid leukemia.

^a Statistically significant.

compare rates directly, however, because we had no accurate census counts for the Hmong population in Minnesota. We also probably underestimated the true PIRs of cancer in the Minnesota Hmong population, because our analyses were limited to surnames specific to the Hmong population.

The results of this study may have implications for the early detection of certain cancers in this population. For example, an increased incidence of cervical cancer might be addressed, at least in part, by targeting culturally sensitive screening programs in the Hmong population. Furthermore, due to the markedly increased incidence of nasopharyngeal cancer, it will be important for physicians to consider possible ma-

lignancy when evaluating Hmong patients for signs and symptoms involving the ear, nose, or throat.¹⁹ The Hmong residing in Minnesota have maintained many aspects of their culture and primarily are agrarian. Therefore, as lifestyles of the Hmong change due to Western influences, their cancer incidence rates (especially among descendants) also may change to more closely reflect the cancer incidence rates of the native U.S. population (for example, both colorectal cancer and breast cancer incidence rates in the U.S.-born Japanese population now approach the rates in the non-Hispanic white population).^{20,21}

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